Sanjay Siva Kumar Surampudi

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LinkedIn | GitHub | Portfolio

PROFESSIONAL SUMMARY

Aspiring VLSI Design and Verification Engineer with a strong foundation in digital IC design, Verilog HDL, semiconductor fundamentals, embedded systems, and IoT.

SKILLS AND PROFICIENCIES

Languages: Verilog, SystemVerilog, C, C++, Python, Java
Tools: Xilinx Vivado, Gowin, EDA Playground, Git, Linux, Docker
Software: Eagle (PCB Design), Proteus, MS Excel, Power BI, PSpice, MATLAB, AutoCAD
Soft Skills: Team collaboration and communication, problem-solving and debugging mindset, logical thinking and precision, time management and task prioritization

WORK EXPERIENCE

Technical Hub Pvt Ltd — **May 2026 – June 2026**
VLSI RTL Design Intern — *Surampalem, Andhra Pradesh*

- Designed, simulated, and debugged 10+ Verilog RTL modules, strengthening expertise in digital logic designs,finite-state machines and combinational and sequential circuits .
- Developed and RTL-to-GDSII Automation Platform and Browser-based Verilog IDE that integrated code editing, simulation, synthesis, and physical design into a single workflow, reducing manual design setup time by approximately 70%.

Eduexpose — **May 2026 – June 2026**
IoT and Embedded Systems Intern — *Hyderabad, Telangana*

- Built embedded applications using ESP32 and Arduino, interfaced 5+ sensors, and implemented real-time data acquisition for IoT-based systems.
- Built a Real-time Data Monitoring System that updated sensor readings every 2 seconds and displayed live data through a cloud-based web dashboard for remote monitoring.

PROJECTS

RTL-to-GDSII Automation Platform — **2026**
Icarus Verilog, SKY130, OpenLane, Yosys, KLayout, HTML, CSS, Cloudflare, ngrok

- Automated the complete RTL-to-GDSII Automation Platform by integrating OpenLane, Yosys, Magic, Netgen, and KLayout into a web platform, reducing manual execution from 15+ steps to a single automated workflow.

Browser-Based Verilog IDE — **2026**
HTML, CSS, Icarus Verilog, Yosys, Docker

- Built a browser-based Verilog IDE supporting code editing, compilation, simulation, and waveform visualization, successfully validating more than 30 Verilog designs.

Long-Distance Offline Communication Using LoRa — **2026**
Arduino Uno, LoRa SX1278, Neo-6M GPS, C/C++, Arduino IDE

- Engineered a LoRa-based emergency communication system providing wireless communication over distances of up to 15 km with real-time GPS location sharing in areas without cellular connectivity.

Real-time Data Monitoring System — **2026**
ESP32, IoT, Firebase, HTML/CSS, JavaScript

- Designed an IoT monitoring platform using ESP32 and Firebase that refreshed sensor data every 2 seconds and displayed live readings through a responsive web dashboard.

ACHIEVEMENTS

- **Coding Profiles:**Solved more than 125 HDLBits exercises and over 500 programming problems across LeetCode and CodeChef.
- **Athletics:** Gold Medal in State-level Taekwondo; Silver Medal in National-level Taekwondo.
- **HackerRank Badges:** Earned 3-star proficiency badges in Problem Solving and C.

CERTIFICATIONS

- **Cadence:** Semiconductor 101, Digital IC Design, Verilog Language and Applications
- **Microsoft:** Microsoft Excel Certified
- **HackerRank:** Problem Solving Certified

EDUCATION

Aditya University, Surampalem — **August 2024 – Expected 2028**
Bachelor of Technology, Electronics and Communication Engineering — *CGPA: 8.36*